

01286326 THEORY OF COMPUTATION ASSIGNMENT 1

PROBLEM 1

Suppose M_1 is a finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0, 1\}$
- δ is described as

	0	1
q_0	q_1	q_3
q_1	q_2	q_1
q_2	q_4	q_1
q_3	q_3	q_3
q_4	q_4	q_2

- q_0 is the initial state,
- $F = \{q_1, q_4\}$

- 1.1) Draw a state diagram of machine M_1 .
- 1.2) Write the computation path of M_1 on the string 010011. Is this string accepted by M_1 ?
- 1.3) Find all the strings of length no greater than 4 accepted by M_1 .

PROBLEM 2

Let L_1 and L_2 be the following languages on the alphabet $\Sigma = \{0, 1\}$:

$$L_1 = \{w \in \Sigma^* \mid w \text{ contains an even number of occurrences of } 1\}$$

$$L_2 = \{w \in \Sigma^* \mid w \text{ contains no adjacent occurrences of } 1\}.$$

- 2.1 Design (deterministic) finite automata M_1 and M_2 that recognize L_1 and L_2 , respectively.
- 2.2 Construct a product automaton of M_1 and M_2 that recognizes the language $L_1 \cap L_2$.
- 2.3 Construct a product automaton of M_1 and M_2 that recognizes the language $L_1 \cup L_2$.

PROBLEM 3

3.1 Design a (deterministic) finite automaton that recognizes the following language L on the alphabet $\Sigma = \{0, 1\}$:

$$L = \{0011, 1010, 1110, 0100, 111, 01\}.$$

3.2 Design a (deterministic) finite automaton that recognizes the following language L' on the alphabet $\Sigma = \{0, 1\}$:

$$L' = \{w \in \Sigma^* \mid w \text{ is a binary representation of a non-negative integer divisible by } 3\}.$$

PROBLEM 4

Suppose M_2 is a nondeterministic finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

- $Q = \{p_0, p_1, p_2, p_3, p_4\}$
- $\Sigma = \{0, 1\}$
- δ is described as

	ϵ	0	1
p_0	$\{p_1\}$	$\{p_4\}$	$\{\}$
p_1	$\{\}$	$\{p_1\}$	$\{p_1, p_2\}$
p_2	$\{\}$	$\{\}$	$\{p_3\}$
p_3	$\{\}$	$\{\}$	$\{\}$
p_4	$\{\}$	$\{p_4\}$	$\{\}$

- p_0 is the initial state,
- $F = \{p_3, p_4\}$

4.1) Draw a state diagram of machine M_2 .

4.2) Write a computation tree of M_2 on the string 01011. Is this string accepted by M_2 ?

4.3) Apply subset construction to construct a (deterministic) finite automaton that is equivalent to M_2 .

PROBLEM 5

Construct a nondeterministic finite automaton recognizing the following language $\Sigma^* - (L_1 L_2)$ where L_1 and L_2 are languages on the alphabet $\Sigma = \{0, 1\}$ given as follows:

$$L_1 = \{w \in \Sigma^* \mid w \text{ contains an even number of occurrences of } 1\}$$

$$L_2 = \{w \in \Sigma^* \mid w \text{ contains no adjacent occurrences of } 1\}.$$

PROBLEM 1

Suppose M_1 is a finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0, 1\}$
- δ is described as

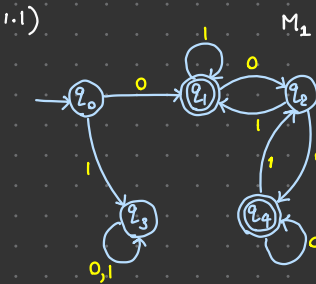
	0	1
q_0	q_1	q_3
q_1	q_2	q_1
q_2	q_4	q_1
q_3	q_3	q_3
q_4	q_4	q_2

- q_0 is the initial state,
- $F = \{q_1, q_4\}$

1.1) Draw a state diagram for machine M_1 .

1.2) Write the computation path of M_1 on the string 010011. Is this string accepted by M_1 ?

1.3) Find all the strings of length no greater than 4 accepted by M_1 .



1.2) 010011

$q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_2 \xrightarrow{0} q_4 \xrightarrow{0} q_2 \xrightarrow{1} q_1$

1.3) strings of length no greater than 4 accepted by M_1

$= \{ \epsilon, 01, 000, 001, 011, 0000, 0111, 0101, 0011 \}$

PROBLEM 2

Let L_1 and L_2 be the following languages on the alphabet $\Sigma = \{0, 1\}$:

$$L_1 = \{w \in \Sigma^* \mid w \text{ contains an even number of occurrences of } 1\}$$

$$L_2 = \{w \in \Sigma^* \mid w \text{ contains no adjacent occurrences of } 1\}.$$

2.1 Design (deterministic) finite automata M_1 and M_2 that recognize L_1 and L_2 , respectively.

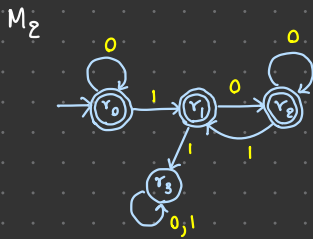
2.2 Construct a product automaton of M_1 and M_2 that recognizes the language $L_1 \cap L_2$.

2.3 Construct a product automaton of M_1 and M_2 that recognizes the language $L_1 \cup L_2$.

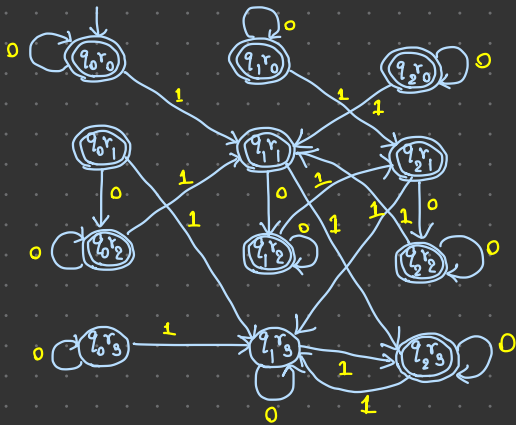
2.1) $L_1 = \{ \epsilon, 01, 110, 101, 0110, 0101, 1010, \dots, 1111, 11110, 01111, \dots \}$



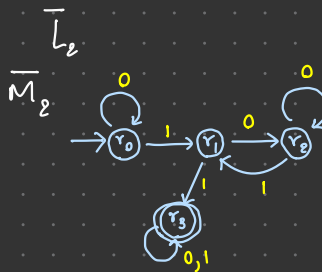
$L_2 = \{ 0, 1, 00, 01, 10, 000, 001, 010, 100, 0000, 1000, 0100, 0010, 0001, 1001, 1010, 0101, \dots \}$



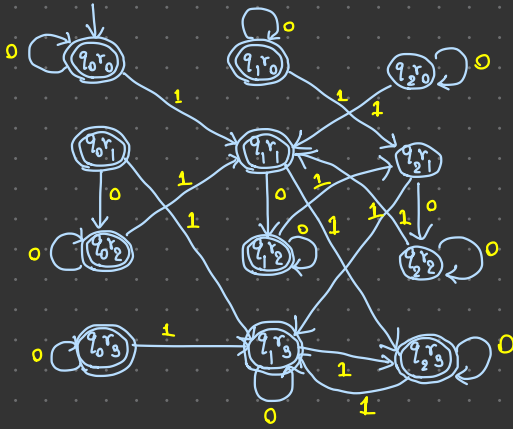
2.3) $L_1 \cup L_2$



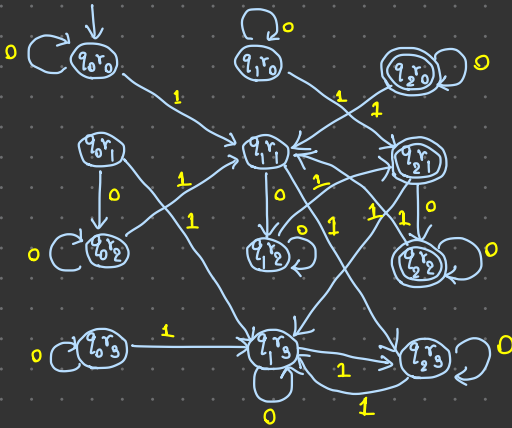
2.2) $L_1 \cap L_2 = \overline{\overline{L_1} \cup \overline{L_2}}$



$$\overline{L_1} \cup \overline{L_2}$$



$$\overline{\overline{L_1} \cup \overline{L_2}} = L_1 \cap L_2$$



PROBLEM 3

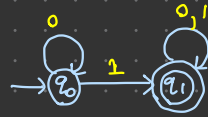
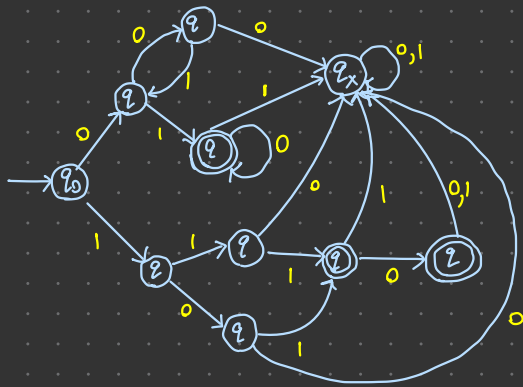
3.1 Design a (deterministic) finite automaton that recognizes the following language L on the alphabet $\Sigma = \{0,1\}$:

$$L = \{0011, 1010, 1110, 0100, 111, 01\}.$$

3.2 Design a (deterministic) finite automaton that recognizes the following language L' on the alphabet $\Sigma = \{0,1\}$:

$$L' = \{w \in \Sigma^* \mid w \text{ is a binary representation of a non-negative integer divisible by 3}\}.$$

3.1) $L = \{0011, 1010, 1110, 0100, 111, 01\}$



$$0011 \Rightarrow \rightarrow q_0 \xrightarrow{0} q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_1 \xrightarrow{1} q_1$$

$$1010 \Rightarrow \rightarrow q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_1 \xrightarrow{1} q_1 \xrightarrow{0} q_1$$

$$1110 \Rightarrow \rightarrow q_0 \xrightarrow{1} q_1 \xrightarrow{1} q_1 \xrightarrow{1} q_1 \xrightarrow{0} q_1$$

$$0100 \Rightarrow \rightarrow q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_1 \xrightarrow{0} q_1$$

$$111 \Rightarrow \rightarrow q_0 \xrightarrow{1} q_1 \xrightarrow{1} q_1 \xrightarrow{1} q_1$$

$$01 \Rightarrow \rightarrow q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_1$$

3.2) $L' = \{11, 110, 1001, 1100, 1111, \dots\}$

PROBLEM 4

Suppose M_2 is a nondeterministic finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

- $Q = \{p_0, p_1, p_2, p_3, p_4\}$
- $\Sigma = \{0, 1\}$
- δ is described as

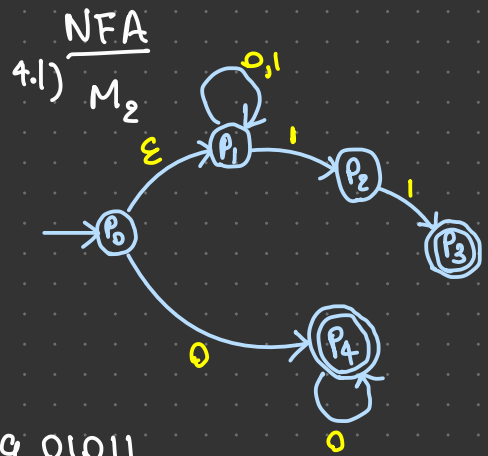
	ϵ	0	1
p_0	$\{p_1\}$	$\{p_4\}$	$\emptyset$
p_1	$\emptyset$	$\{p_1\}$	$\{p_1, p_2\}$
p_2	$\emptyset$	$\emptyset$	$\{p_3\}$
p_3	$\emptyset$	$\emptyset$	$\emptyset$
p_4	$\emptyset$	$\{p_4\}$	$\emptyset$

- p_0 is the initial state,
- $F = \{p_3, p_4\}$

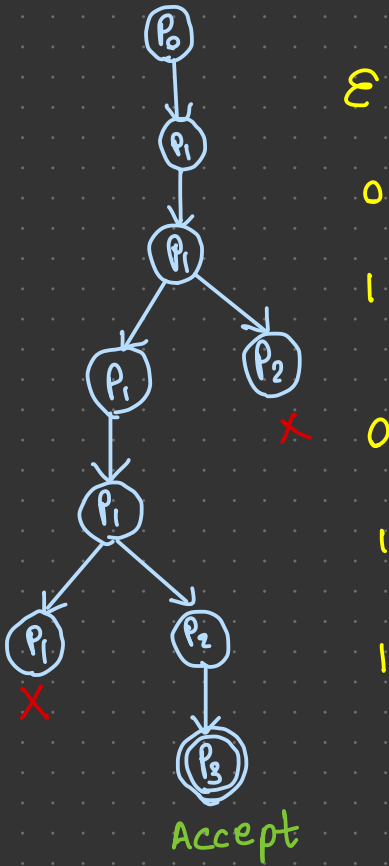
4.1) Draw a state diagram of machine M_2 .

4.2) Write a computation tree of M_2 on the string 01011. Is this string accepted by M_2 ?

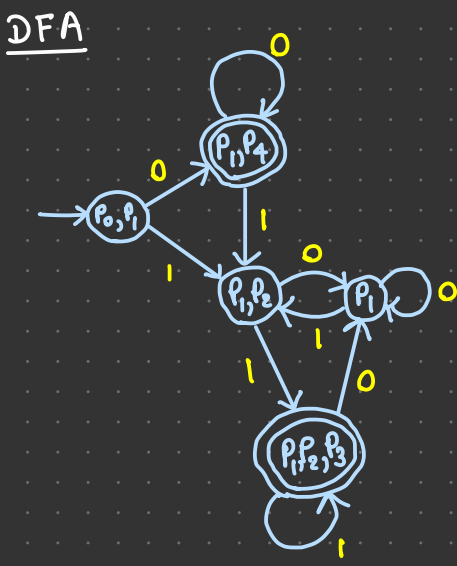
4.3) Apply subset construction to construct a (deterministic) finite automaton that is equivalent to M_2 .



4.2) Computation tree of M_2 for the string 01011



DFA State	0	1
$\{p_0, p_1\}$	$\{p_1, p_4\}$	$\{p_1, p_2\}$
$\{p_1, p_2\}$	p_1	$\{p_1, p_2, p_3\}$
$\{p_1, p_4\}$	$\{p_1, p_4\}$	$\{p_1, p_2\}$
p_1	p_1	$\{p_1, p_2\}$
$\{p_1, p_2, p_3\}$	p_1	$\{p_1, p_2, p_3\}$



PROBLEM 5

Construct a **nondeterministic** finite automaton recognizing the following language $\Sigma^* - (L_1 L_2)$ where L_1 and L_2 are languages on the alphabet $\Sigma = \{0,1\}$ given as follows:

$$L_1 = \{w \in \Sigma^* \mid w \text{ contains an even number of occurrences of } 1\}$$

$$L_2 = \{w \in \Sigma^* \mid w \text{ contains no adjacent occurrences of } 1\}.$$

$$L_1 = \{\epsilon, 0, 011, 110, 101, 0110, 0101, 1010, \dots, 1111, 11110, 01111, \dots\}$$

$$L_2 = \{\epsilon, 1, 00, 01, 10, 000, 001, 010, 100, 0000, 1000, 0100, 0010, 0001, 1001, 1010, 0101, \dots\}$$

$$L_1 L_2$$

